

# Eric Pedley

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## Education

**University of California, Irvine**

GPA: 3.80 | B.S. Computer Science

September 2021 - Spring 2025

## Skills

**Languages:** Python, C++, C, TypeScript, JavaScript, HTML, CSS, Java, Assembly

**Frameworks/Libraries:** OpenCV, PyTorch, Numpy, JAX, TensorFlow, ROS2, Ultralytics

**Other Technologies:** Git, Linux (Ubuntu & Arch), Gazebo, AWS, Docker, Microcontrollers, Nvidia Jetson Orin

## Work Experience

**Robotics Contractor | K-SCALE LABS**

Menlo Park, CA | August - September 2025

- Made software allowing teleoperation of a humanoid robot upper-body with a VR headset with custom hardware-accelerated inverse kinematics
- Prototyped RL-based whole-body control using Isaac Lab
- Worked on supporting tasks such as updating/bugfixing Rust firmware to support different actuator configurations and logging.

**Software Development Intern | TAU MOTORS**

Redwood City, CA | June - September 2024

- Added new features to developer tooling to enable faster debugging and cleaner code
- Used nonlinear optimization to implement a new state estimation technique

**Software Development Engineering Intern | AWS S3**

Arlington, VA | June - September 2023

- Prototyped a new feature for S3 to detect duplicate objects
- Used batch processing and streaming to scale service to over 1 billion objects

## Extracurriculars

**Cyclone Labs Software Lead**

October 2024 - Present

- Lead a team in the Abu Dhabi Autonomous Racing League
- Trained a keypoint detection network using synthetic data to detect racing gates

**UCI UAV Forge Perception Subteam Lead**

September 2023 - June 2025

- Lead 7 other students developing of a system for locating ground-based targets and avoiding other drones for the Student Unmanned Aerial Systems competition
- Created a multi-label object detection deep learning architecture from scratch and a synthetic data pipeline
- Implemented algorithms to detect other drones using object detection and particle filters.
- Designed end-to-end software tests in Gazebo sim with Ardupilot SITL

**UCI Rocket Project Avionics Subteam Member**

June 2023 - May 2025

- Designed a camera tracking system to record our rocket and keep a directional telemetry antenna on target
- Used Kalman filters to fuse camera object tracking and GPS/altimeter telemetry data
- Distilled foundational object detection model on scraped images of rockets to run on edge hardware
- Wrote C++ code to read GPS on our onboard telemetry package without blocking other sensors

## Projects

**Self-Driving Car Pose Graph Optimization** 

GTSAM, FACTOR GRAPHS

- Toy project to learn factor graph optimization where I used visual odometry and factor graphs to estimate the location of a car in the TUM 4Seasons dataset.

**Pose Cloning with Inverse Kinematics** 

PYTHON, ROS2

- Wrote a program to track a person's pose and map it onto a model of Boston Dynamics' Atlas robot using inverse kinematics.

**Reinforcement Learning Quadcopter Control** 

REINFORCEMENT LEARNING

- Used PPO to train a policy that allows a quadcopter to navigate to waypoints.